

DATA STRUCTURES AND ALGORITHMS –
3 (25CS2103E)

SMART RECIPES

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CSE





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Abstract



SMART RECIPES is a recipe recommendation and text-processing system that helps users find suitable recipes based on available ingredients and search queries. Unlike basic recipe applications, it uses advanced Data Structures and Algorithms as a core part of the system.

The system supports pattern matching using KMP, fuzzy searching using Levenshtein Distance, and recipe similarity using Suffix Arrays and LCP. It also provides ingredient-based recipe recommendations and ranks recipes based on the number of matching ingredients.

The project helps users find recipes more efficiently, handle spelling mistakes, compare recipe information, save time, and reduce food wastage.



Introduction



Finding suitable recipes can become difficult when users have many ingredients but do not know what they can prepare. Traditional recipe applications mainly provide simple keyword searching and recipe browsing.

Smart Recipes provides a console-based solution where users can enter available ingredients, search for recipes, perform fuzzy searches when there are spelling mistakes, and compare recipes using text similarity.

The main feature of the project is that DSA-3 algorithms are used as the core processing engine, rather than only for demonstration.

Literature Survey

Traditional recipe systems mainly provide features such as:

Recipe browsing

Keyword-based searching

Ingredient lists

Cooking instructions

Category-based filtering

.

Smart Recipes focuses on:

KMP-based pattern matching

Levenshtein Distance for fuzzy searching

Suffix Array and LCP for text similarity

DSA-based ingredient matching

Sorting and ranking of recommended recipes



Problem Statement and Objectives

Problem Statement:

Users may have ingredients available at home but may not know what recipes can be prepared. Searching manually for recipes can take time, and simple search systems may fail when users enter spelling mistakes. Basic recipe applications may also not provide efficient text similarity analysis or algorithm-based recommendations.

Objectives:

Centralize recipe and ingredient information.

Search recipes efficiently using pattern matching.

Support fuzzy searching for incorrect spellings.

Recommend recipes based on available ingredients.

Compare recipe documents using text similarity.

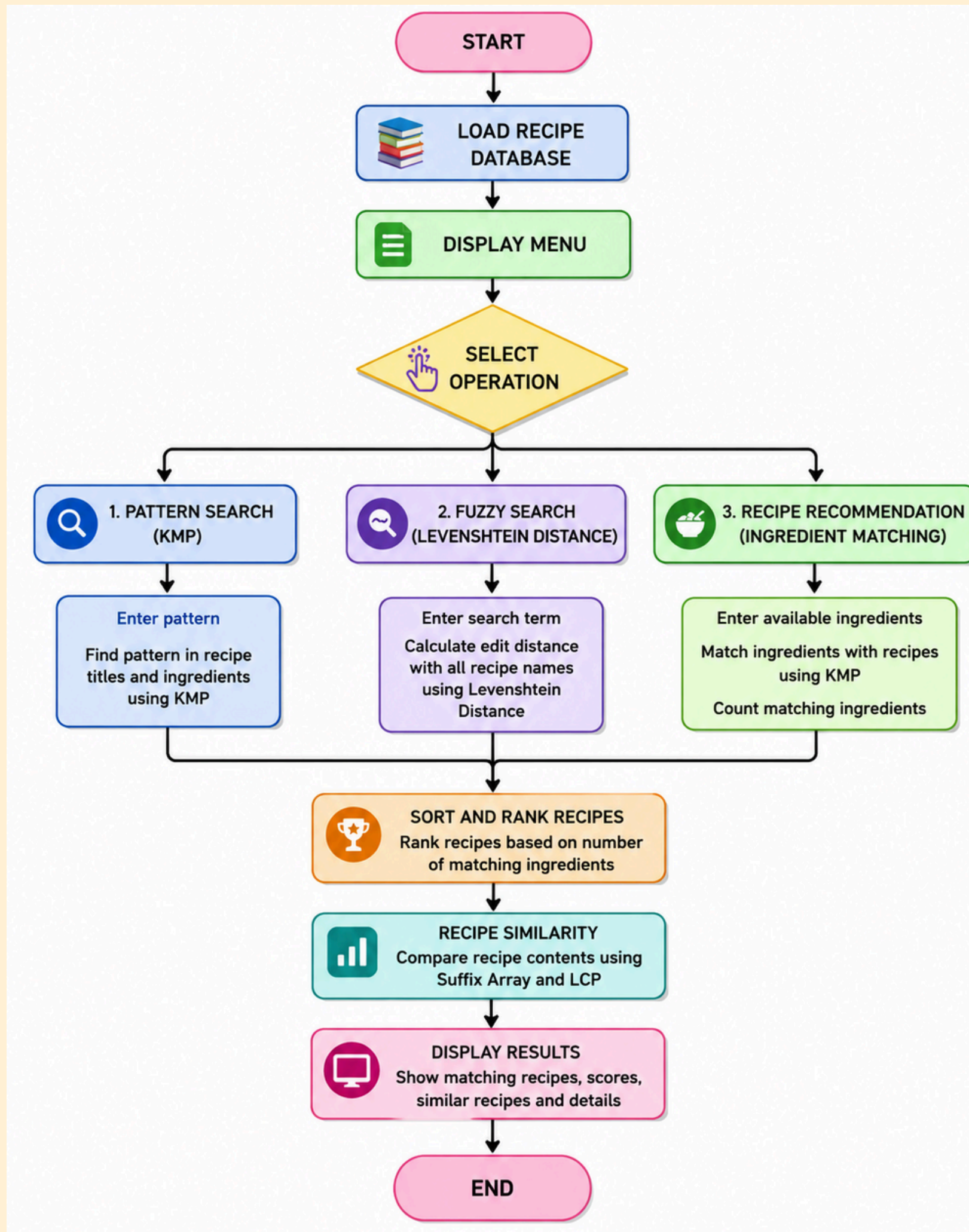
Rank recipes based on matching ingredients.

Reduce manual searching time.

Reduce food wastage by suggesting recipes using available ingredients.



Flow Chart



Results

The proposed system provides:

Efficient pattern searching using KMP.

Fuzzy search for spelling mistakes using Levenshtein Distance.

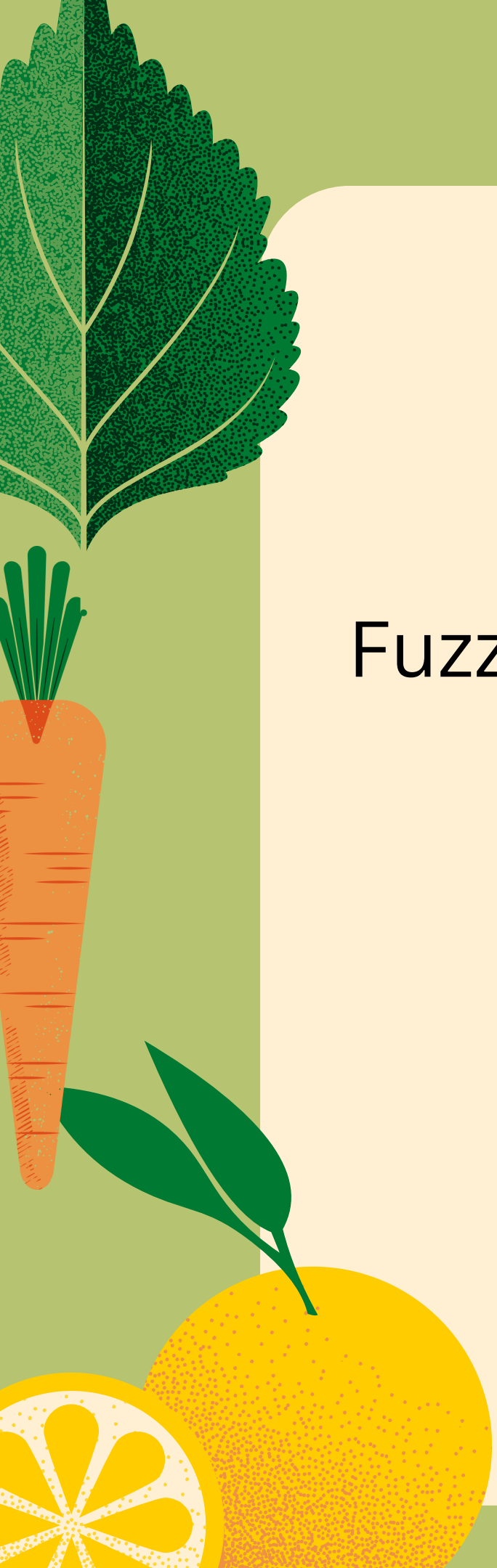
Recipe similarity analysis using Suffix Arrays and LCP.

Ingredient-based recipe recommendations.

Ranking of recipes based on matching ingredients.

Faster and more organized recipe searching.

Reduced manual effort in finding suitable recipes.



Hardware and Software Requirements

Hardware Requirements

- Computer / Laptop
- Minimum 4 GB RAM
- Minimum 1 GB free storage
- Internet connection for GitHub repository

Software Requirements

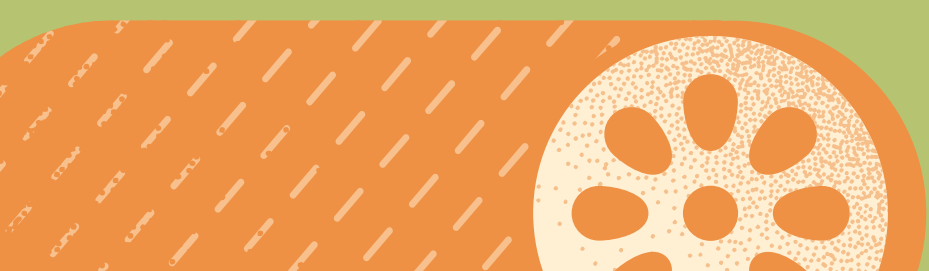
- programming Language: Java
- IDE: Visual Studio Code
- JDK: JDK 17 or above
- Version Control: Git and GitHub
- Database: Text file for the current partial implementation
 - DSA: KMP, Dynamic Programming, Levenshtein Distance, Suffix Array, LCP, Searching and Sorting



Conclusion

Smart Recipes provides an efficient solution for recipe searching and recommendation. Its major advantage is that advanced DSA-3 algorithms are used as a core part of the system rather than only for demonstration. KMP enables efficient pattern searching, Levenshtein Distance handles spelling errors through Dynamic Programming, and Suffix Arrays with LCP support recipe text similarity analysis.

These algorithms help the system provide better searching, fuzzy matching, similarity analysis, and ingredient-based recipe recommendations while reducing the user's manual effort.



Future Scope

1. AI-Based Recipe Recommendations

Recommend recipes by analyzing user preferences, ingredients, and previous searches.

2. Substitute Ingredient Suggestions

Suggest alternative ingredients when a required ingredient is unavailable.

3. Advanced Filtering

Add filters such as:

Vegetarian

Non-vegetarian

4. User Favorites and Ratings

Allow users to save favorite recipes and rate them.

5. Shopping List Generation

Automatically create a list of missing ingredients.

6. Web or Mobile Application

Convert the console-based project into a web or mobile application.

7. Advanced multi-pattern matching

Graph-based recommendation relationships



THANK YOU